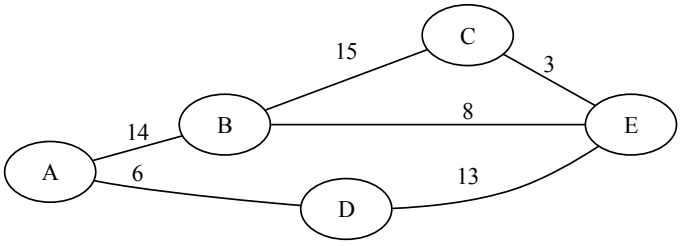
Name _____

Quiz 6

1. Consider the following weighted, undirected graph G.

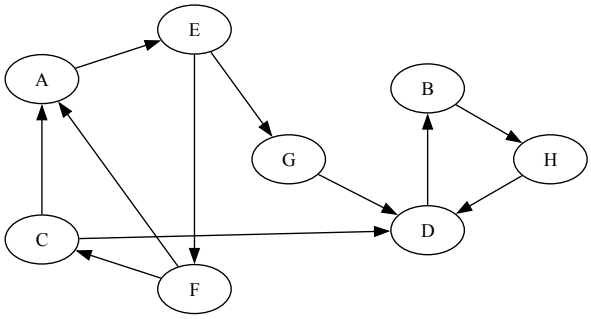


Run WhateverFirstSearch(C) using the algorithm below. Initially, all vertices are unmarked and the bag is empty. The bag is a priority queue: remove the entry with the smallest priority first. When a neighbor is added, its priority is the sum of all edge weights along the path used to reach this entry, including the edge used to add it. The first node added has priority 0. If bag entries have the same priority, remove the one with the alphabetically earliest vertex label first.

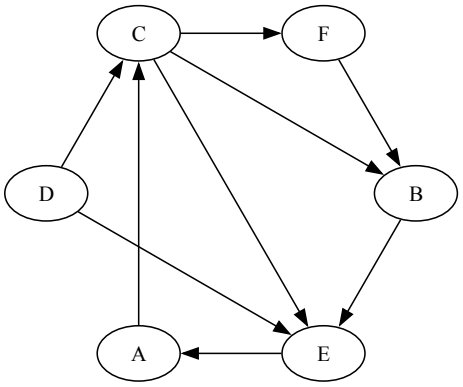
```
WhateverFirstSearch(s):
put s into the bag
while the bag is not empty:
    take v from the bag
    if v is unmarked:
        mark v
        for each neighbor w of v:
            put w into the bag
```

In what order are the vertices marked? Write the vertex order below.

2. What are the strongly connected components in the graph below?



3. Give the preorder values for all nodes in the following graph. The *DFS* and *DFSAll* algorithms are reproduced below if you need them. Assume that both process vertices in alphabetical order.



$A.pre =$ ___, $B.pre =$ ___, $C.pre =$ ___, $D.pre =$ ___, $E.pre =$ ___, $F.pre =$ ___

```
DFSAll(G):
  clock = 0
  unmark all vertices
  for each vertex v:
      if v is unmarked:
          clock = DFS(v, clock)
```

```
DFS(v, clock):
  mark v
  clock = clock + 1
  v.pre = clock
  for each neighbor w of v:
      if w is unmarked:
          w.parent = v
          clock = DFS(w, clock)
  clock = clock + 1
  v.post = clock
  return clock
```